

|                            |                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(c) Change of state</b> |                                                                                                                                                                |
| <b>Students should:</b>    |                                                                                                                                                                |
| <b>5.8P</b>                | <b>explain why heating a system will change the energy stored within the system and raise its temperature or produce changes of state</b>                      |
| <b>5.9P</b>                | <b>describe the changes that occur when a solid melts to form a liquid, and when a liquid evaporates or boils to form a gas</b>                                |
| <b>5.10P</b>               | <b>describe the arrangement and motion of particles in solids, liquids and gases</b>                                                                           |
| <b>5.11P</b>               | <b><i>practical: obtain a temperature–time graph to show the constant temperature during a change of state</i></b>                                             |
| <b>5.12P</b>               | <b>know that specific heat capacity is the energy required to change the temperature of an object by one degree Celsius per kilogram of mass (J/kg °C)</b>     |
| <b>5.13P</b>               | <b>use the equation:</b><br><b>change in thermal energy = mass × specific heat capacity × change in temperature</b><br>$\Delta Q = m \times c \times \Delta T$ |
| <b>5.14P</b>               | <b><i>practical: investigate the specific heat capacity of materials including water and some solids</i></b>                                                   |